

Linear Partial differential eq^{ns} of the first order :-

A d-eqn involving Partial derivatives
 p & q only & no higher order derivatives
is called a first order partial d-eqn.

If p and q occur in the first degree, it-
is called a linear Partial diff eqⁿ of
first-order, ~~& first degree~~ otherwise it-is
called non-linear Partial d-eqn of first
~~order~~ order.

ex $px + qy^2 = z$ is a linear PDE of first order. and $p^2 + q^2 = 1$ is non-linear.

Lagrange's linear Equation:-

A linear Partial DE of order one involving a dependent variable z and two independent variables x & y

of the form $Pp + Qq = R$.

where P, Q, R are fns of x, y, z is called Lagrange's linear eq^{ns}.

Lagrange's auxiliary eq^{ns} ~~is~~ is $\rightarrow$

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

working Rule to solve Lagrange's linear eq^{ns} $Pp + Qq = R$:-

Step ①

write down the auxiliary eq^{ns}

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

(2) Solve the auxiliary eqs by the method of grouping and the method of multipliers ~~to get~~ or both to get the independent solutions $u = a$ & $v = b$, where a & b are arbitrary constants.

(3) Then $\phi(u, v) = 0$ or $u = f(v)$ is the general soln of the eqn $Pp + Qq = R$.

to solve $\frac{dx}{P(x, y, z)} = \frac{dy}{Q(x, y, z)} = \frac{dz}{R(x, y, z)}$.

(i) Method of grouping :-

taking two of the eqns i.e

$$\frac{dx}{P} = \frac{dy}{Q} \quad \text{or} \quad \frac{dy}{Q} = \frac{dz}{R} \quad \text{or}$$

$$\frac{dx}{P} = \frac{dz}{R}.$$

are directly solvable to get solns

$$u(x, y) = \text{constant} \quad \text{or} \quad v(x, y) = \text{constant}$$

$$\text{or} \quad w(x, y) = \text{constant}.$$

These give the complete soln of ①
we can write the soln as

$$u = f(v) \quad \text{or} \quad v = f(u) \quad \text{or}$$

$$\phi(u, v) = 0.$$

~~ii)~~

ii) Method of Multiplier's :-

$$\text{if we have } \frac{dx}{p} = \frac{dy}{q} = \frac{dz}{r}$$

take multipliers l, m, n , so that -

$$\text{each ratio } \frac{l dx + m dy + n dz}{lp + mq + nr}$$

where $lp + mq + nr = 0$ then

$$l dx + m dy + n dz = 0. \quad \text{integrating}$$

$$l dx + m dy + n dz = 0 \quad \text{we get -}$$

$$u(x, y, z) = C_1,$$

Similarly we get another soln

$$v(x, y, z) = C_2.$$

So we get the complete soln.

Q ① Solve $yz - xp = z$

Sol Here $P = -x$, $Q = y$, $R = z$

Lagrange's auxiliary eqn is

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{-x} = \frac{dy}{y} = \frac{dz}{z}$$

$\Rightarrow$ taking first two fractions

$$\frac{dx}{-x} = \frac{dy}{y}$$

$$\Rightarrow -\log x = \log y + \log C_1$$

$$\Rightarrow -\log x - \log y = \log C_1$$

$$\Rightarrow \log xy = \log C_1$$

$$\Rightarrow xy = C_1 \text{ ——— ①}$$

taking last two fractions

$$\frac{dy}{y} = \frac{dz}{z}$$

$$\log y = \log z + \log C_2$$

$$y = zC_2 \Rightarrow C_2 = \frac{y}{z} \text{ ——— ②}$$

from (1) & (2), the complete solⁿ is
 $f\left(xy, \frac{y}{z}\right) = 0.$

2) Solve $y^2 p - xy q = x(z - 2y)$

Sol Lagrange's auxiliary eqn is

$$\frac{dx}{y^2} = \frac{dy}{-xy} = \frac{dz}{x(z-2y)}$$

→ taking first two fractions

$$\frac{dx}{y^2} = \frac{dy}{-xy}$$

$$-x dx = y dy$$

$$\Rightarrow -x dx - y dy = 0$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = C_1 \quad \text{--- (1)}$$

taking last two

$$\frac{dy}{-xy} = \frac{dz}{x(z-2y)}$$

$$\Rightarrow \cancel{\log(z-2y)}$$

$$-\frac{dy}{y} = \frac{dz}{(z-2y)}$$

$$\Rightarrow -(z-2y)dy = ydz$$

$$\Rightarrow -zdy + 2ydy = ydz$$

$$\Rightarrow 2ydy = ydz + zdy$$

$$\Rightarrow 2 \int y dy = \int d(yz)$$

$$\Rightarrow \cancel{2} \cdot \frac{y^2}{\cancel{2}} = yz + c_2$$

$$\Rightarrow y^2 - yz = c_2$$

hence the complete soln is

$$\phi(x^2+y^2, y^2-yz) = 0.$$

$$\text{or } x^2+y^2 = f(y^2-yz).$$

$$\textcircled{3} \text{ solve } y^2 z p + x^2 z q = y^2 x$$